

KIDS DEVELOPMENTAL PROTOTYPE
NOT CHILD-VALIDATED
NOT PUBLICATION-READY

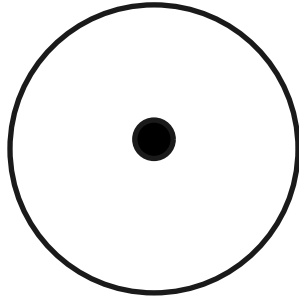
ONE TAP / INPUT → RESPONSE — KIDS-ELEM2

Age guide: Grades 3–5/6

Review prototype · style-neutral figures · not child-validated

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S01 — Cross-layer path (OBSERVE)



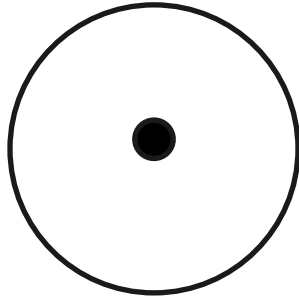
Follow one tap with Mira across layers you can name on paper: human intent → input hardware → operating-system event → application handler → optional network → output (pixels or sound) → your perception. Annotate the shared map. Circle steps you truly observed. Put a question mark on steps you only infer. A map is a model — useful, incomplete, and revisable. If two classmates mark different question marks, that is good science, not a fight. Compare notes before you “fix” anything. Leave one blank margin note titled Still invisible to us.

Try it: Annotate cross-layer map; mark observe vs infer.

Talk together: Keep discussion concrete; shared figure is a tool, not a quiz key.

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S02 — CPU, memory, storage (NAME)

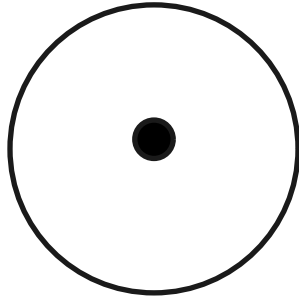


Bolt deals three role cards. CPU: runs instructions right now. Memory (working space): holds state the program is using this moment. Storage (keep-box): keeps durable data for later. Sort classroom examples: the instructions being followed, today's draft text still open, yesterday's saved file. Say the roles aloud. We describe jobs machines do. A bigger number on an advertisement does not tell the whole story — ask what was actually measured. If a label sounds like an ad for “brain speed,” rewrite it as a job the machine can do.

Try it: Sort role cards for CPU / memory / storage.
Talk together: Roles, not marketing numbers.

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S03 — OS and app events (NAME)



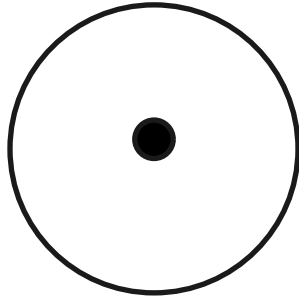
A tap becomes an event. The operating system notices input-hardware activity and delivers an event toward the app that should respond. Inside the app, an event loop can dispatch a handler that updates state and asks for a new frame of output. Draw one chain on a single page: tap → OS event → app handler → updated state → screen or sound. This is a classroom model — useful, incomplete, and revisable. If a word is new, define it once with a pointing example from the chain you drew. Then explain the same chain in your own words.

Try it: Draw event-handling sequence.

Talk together: Keep the chain on one page; define new words once with pointing.

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S04 — Packets or not? (PREDICT)



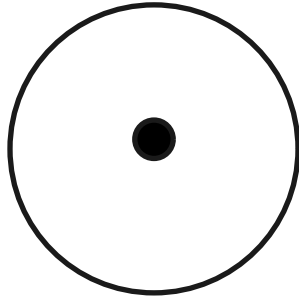
Ping asks: does your action need packets — little addressed message pieces that can travel on a network? Predict first, in writing. Then imagine airplane mode or radio off: what would you observe if the action were local only? What would fail, hang, or stay blank if it needed the network? Fill three boxes: prediction → planned observation → result. Local vs network is a question for evidence, not a slogan you memorize. If you cannot test safely today, keep the prediction and write why the test waited.

Try it: Predict packet need; plan an observation if radio is off.

Talk together: Prediction before trial.

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S05 — Latency: feel vs measure (MEASURE)

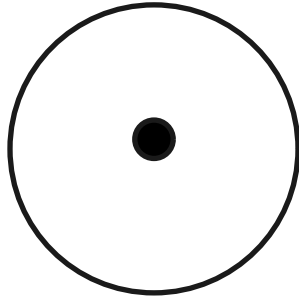


Latency means waiting time somewhere on a path. Felt delay is what your body notices; measured delay is what a timer roughly shows. Use a classroom stopwatch on a remote refresh. Record three marks if you can: start, first visible change, and content “done enough.” Separate felt delay from guessed cause — slow Wi-Fi, a busy CPU, and a far server can feel similar until you gather evidence. Rough numbers beat confident myths. Write units (seconds) beside every mark.

Try it: Rough stopwatch measure of a remote refresh.
Talk together: Classroom timer OK; rough is honest.

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S06 — Observation vs inference (EXPLAIN)



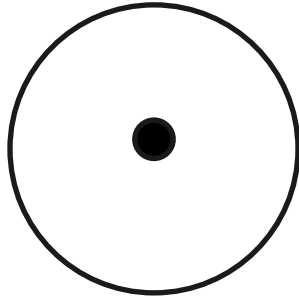
Two columns only: Observation | Inference. Example observation: “Highlight appeared in under a second.” Example inference: “Therefore no packets were sent.” Mark every inference with a ? until tested. Systems literacy starts when we stop pretending every story we tell ourselves is a measurement. Trade papers with a partner and circle any inference dressed up as a fact. Add a third tiny box if needed: Still don’t know — and keep it empty of fake certainty.

Try it: Two-column notes with uncertainty marks.

Talk together: Practice the columns before you argue causes.

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S07 — Security vs privacy (SECURE)



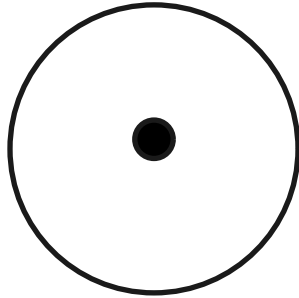
Shield separates two ideas that often get mashed together. Security: controls that protect systems and accounts — lock screen, updates, permission prompts. Privacy: choices about who can see or keep your information. List one security control and one privacy choice that change a tap path. We may talk about HTTPS as “the path tries to keep the message harder for strangers to read on the way.” We learn what tools protect people — we do not practice breaking into systems. Calm tools beat scary stories.

Try it: List one security control + one privacy choice.

Talk together: Protection tools, not scare stories.

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S08 — Build, test, teach (BUILD)

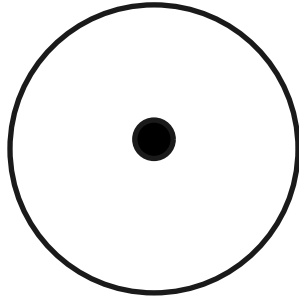


Build a paper protocol for tracing one tap: numbered steps, a failure branch (“if no change, check power / ask a trusted adult”), and a teach-back script for someone younger (short sentences; no stack jargon). Test the protocol once with a partner. Repair one unclear step and date the repair. Your work packet = protocol + teach script + one repair note. Repair is success.

Try it: Paper protocol + younger teach script + one repair.
Talk together: Work packet shows process evidence.

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S09 — Spiral: baby to map (REFLECT)



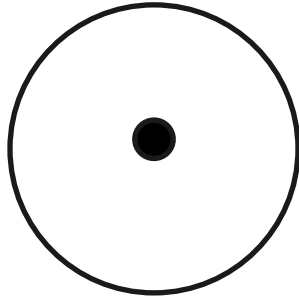
What stays the same from baby LOOK → CHANGE to this cross-layer map? Contingent action and response — something happens, then something answers. What vocabulary was earned later: input, output, event, packet, latency, privacy, security? Spiral learning means the same idea grows tools as learners grow. How would you explain the same idea to someone younger using fewer new words? Write one sentence you would teach a toddler and one you would only teach this class.

Try it: Spiral reflection note.

Talk together: Same idea, fewer new words for younger listeners.

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S10 — EMIT mini-capstone (TEACH)



EMIT loop: Explain one tap path · Measure one wait · Improve one step of your protocol · Teach someone else. With Shield, say what you observed, label what you inferred, and admit what you still do not know. Do not invent evidence to sound finished. Your work packet shows process — not a fake certificate. When you teach, invite questions you cannot answer yet, and write them down for next time.

Try it: Complete EMIT loop; name observed, inferred, and still-unknown.
Talk together: Evidence and uncertainty, spoken calmly.

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Facilitator pointer

Standards + provenance: AUTHOR_NOTES.yaml / TRACEABILITY.yaml

Adult main (provenance only): 82284cd8f41d750ff508cd6ea5bad0a9534d8162

No child validation evidence exists for this prototype.

Claim ceiling: KIDS_REVIEW_PROTOTYPE_COMPLETE